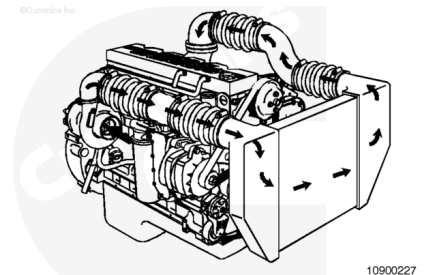


Trainings ▾

General Information

The long-term integrity of the charge-air cooler system is the responsibility of the vehicle and component manufacturers; however, the following can be checked by any Cummins® Authorized Repair Facility.

Note : If the engine experiences turbocharger damage or any other occasion where oil or debris is put into the charge-air cooler, the charge-air cooler must be cleaned.

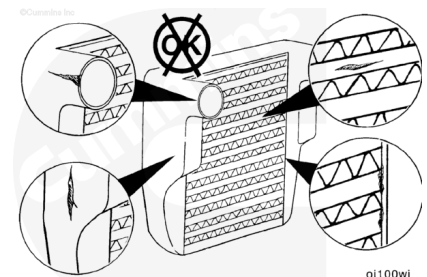


Initial Check

Inspect the charge-air cooler for cracks, holes, and other damage.

Inspect the tubes, fins, and welds for tears, breaks, or other damage. If any damage causes the charge-air cooler to fail the air leak check, the charge-air cooler **must** be replaced.

Inspect the charge-air cooler plumbing for cracks and damage.

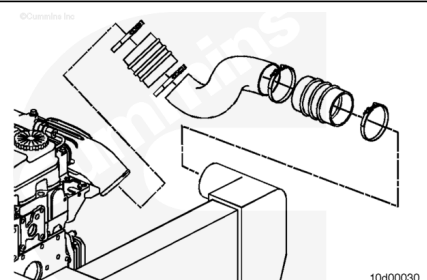


Remove

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠



Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Drain the coolant.

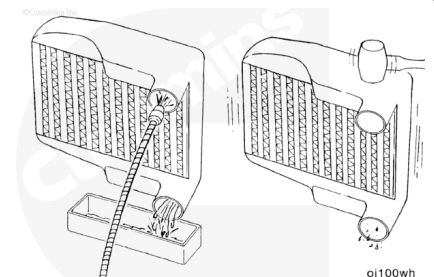
Remove the charge air cooler plumbing and cooler. Refer to original equipment manufacturer (OEM) service manual.

Use compressed air to clean debris from the outside of the charge-air cooler.

Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ CAUTION ⚠

Do not use caustic cleaners to clean the charge air cooler. Damage to the charge air cooler will result.

Flush the charge-air cooler internally with solvent in the opposite direction of normal airflow. Shake the charge-air cooler and lightly tap on the end tanks with a rubber mallet to

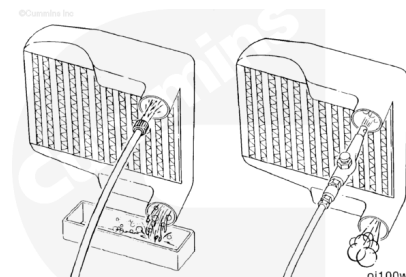
dislodge trapped debris. Continue flushing until all debris or oil is removed (i.e., the water runs clear).

Note : Make sure that the tubes are in the vertical direction when flushing.

If the debris can **not** be totally removed from the charge-air cooler, the charge-air cooler **must** be replaced.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

The charge air cooler must be rinsed, dried, and cleaned of solvent, oil, and debris, or engine damage will result.

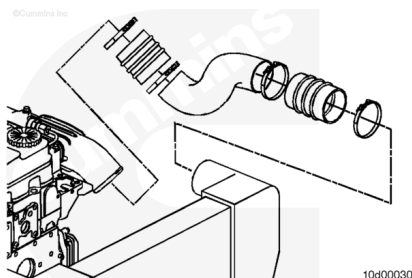
After the charge-air cooler has been thoroughly cleaned of all oil and debris with solvent, wash the charge-air cooler internally with hot, soapy water to remove the remaining solvent. Rinse thoroughly with clean water.

Blow compressed air through the inside of the charge-air cooler in the opposite direction of normal airflow until the charge-air cooler is dry internally.

Install

Install the charge-air cooler.

Install the charge-air cooler plumbing. Refer to the OEM service manual for instructions.



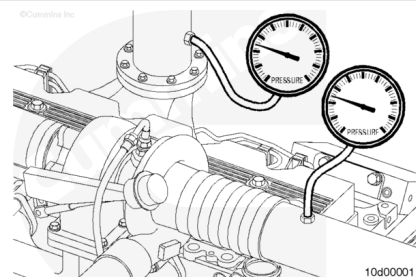
Pressure Test

Install the pressure gauge, Part Number ST-1273, or equivalent, to the fitting in the turbocharger outlet.

Install another pressure gauge, Part Number ST-1273, or equivalent, in the intake manifold.

Operate the engine at rated rpm and load. Record the readings on the two gauges.

If the differential pressure is greater than 21 kPa [3 psi], check the charge-air cooler for plugging. Clean or replace if necessary.



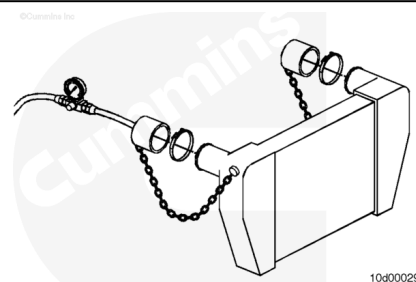
Leak Test

⚠ WARNING ⚠

To reduce the possibility of injury if either plug blows off during the test, secure safety chains on the test plugs to any convenient capscrew on the radiator assembly. This test must not be performed without securely fastened safety chains.

To check the charge-air cooler for cracked tubes or header, remove the inlet and outlet hoses from the cooler. The charge-air cooler does **not** have to be removed from the chassis.

Install a plug or cap over the outlet side of the cooler. Install a pressure gauge and a regulated shop air supply line with a shutoff valve to the inlet side of the cooler.

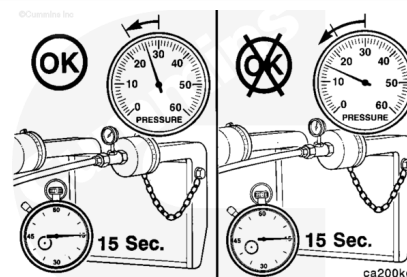


Apply air pressure to the cooler until the pressure gauge reads a steady 207 kPa [30 psi] of air pressure.

Shut off the airflow to the cooler and start a stopwatch at the same time. Record the leakage at 15 seconds.

If the pressure drop is 48 kPa [7 psi] or less in 15 seconds, the cooler is operational.

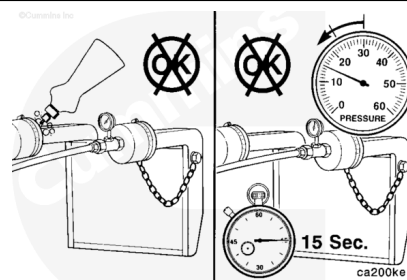
If the pressure drop is greater than 48 kPa [7 psi] in 15 seconds, check all connections again.



Determine if the pressure drop is caused by a leaky charge-air cooler or a leaky connection. Spray soapy water on all hose connections, and watch for bubbles to appear at the location of the leak.

If the pressure drop is caused by a leaky connection, repair the connection and repeat the test. If the leak is within the charge-air cooler, repeat the test to verify the accuracy of the pressure drop measurement. Similar pressure drop readings **must** be obtained in at least three consecutive tests before the reading can be considered accurate.

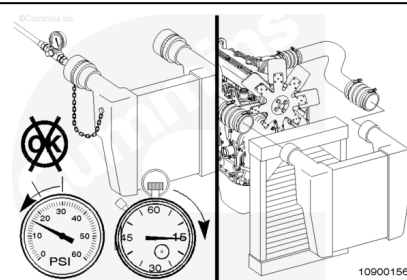
Note : If a charge-air cooler leaks more than 48 kPa [7 psi] in 15 seconds, it will appear as a major leak in a leak tank.



If the pressure drop is greater than 48 kPa [7 psi] in 15 seconds, the charge-air cooler **must** be replaced.

Refer to the OEM service manual for replacement instructions.

Note : Charge-air coolers are not designed to be 100-percent leak free. If the pressure drop is less than 48 kPa [7 psi] in 15 seconds, then the charge-air cooler does not need to be replaced.

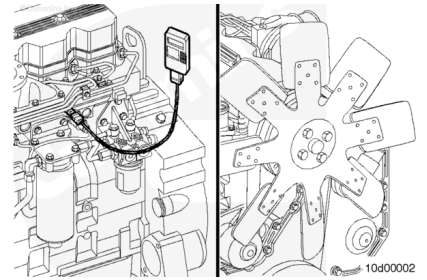


Temperature Differential Test

Install a temperature gauge in the intake manifold.

Lock the fan drive in the ON mode to prevent erratic test results. This can be done by installing a jumper across the temperature switch or supplying shop air to the fan. Refer to the OEM service manual for the lockup procedure.

Note : Some applications have a manual switch that will lock the fan on.



Operate the engine at rated rpm and load. Record the intake manifold temperature.

Measure the ambient temperature at least 2 feet in front of the vehicle.

The maximum temperature differential **must not** be greater than 25°C [77°F].

If the temperature differential is greater than 25°C [77°F], check the charge-air cooler for dirt and debris on the fins, and clean as necessary. If the problem still exists, check the cooler for internal contamination or plugging.

